

Activity 8 Pre-Lab — Markovnikov Hydrohalogenation of Carvone

[Your name] & [partner]
[MM/DD/YY]

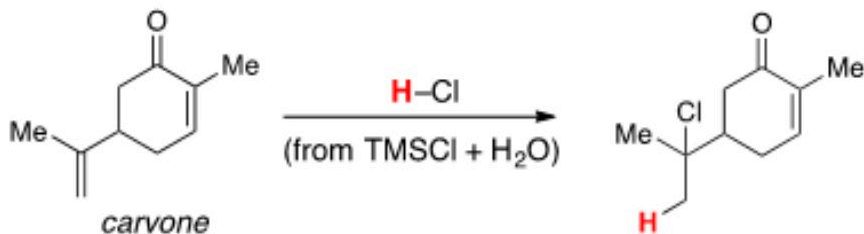
Organic Lab · selective HCl addition to an alkene

■ Black = write this in your notebook ■ Blue = context, don't copy it Copy the header onto every page.

Purpose

Add HCl across an alkene of carvone and show the **chemo- and regioselectivity**: HCl adds **Markovnikov** to the isolated (isopropenyl) C=C — not the conjugated enone C=C — giving the tertiary chloride. Monitor by TLC, then work up and weigh the product.

🔪 DRAW — the reaction (from the procedure)



HCl is generated in situ (TMSCl + H₂O). It

adds Markovnikov to the isolated isopropenyl C=C (H to the CH₂, Cl to the more-substituted carbon → 3° chloride); the conjugated ring enone is unreactive.

Chemical Reagents & Safety Table

CONTEXT ▶ Required: carvone, chlorotrimethylsilane (TMSCl), water. Typical SDS values — cross-check the SDS tile.

Chemical	Role in lab	MW	Amount	Physical properties	GHS pictograms
Carvone	Substrate (limiting)	150.22	2 mL (~1.9 g, ~0.013 mol)	Colorless–pale-yellow liquid; bp 230 °C; d 0.96 g/mL	 irritant
Chlorotrimethylsilane (TMSCl)	HCl source (reacts w/ water)	108.64	~6 mL	Colorless fuming liquid; bp 57 °C; d 0.86 g/mL; flash pt –18 °C	   flammable corrosive irritant
Water	Hydrolyzes TMSCl → HCl	18.02	1 mL	Colorless liquid	— low hazard

FYI ▶ also used (not required in table): sat. NaHCO₃ (quench/neutralize), diethyl ether (extraction), brine, MgSO₄ (drying), 30% EtOAc/hexanes (TLC).

GRAB FIRST: 50-mL RBF + stir bar + clamp + stir plate · plastic pipettes · funnel · dry 10-mL grad cylinder + 250-mL beaker + 50-mL beaker (TMSCI carrier) · beaker w/ ~75 mL sat. NaHCO_3 + aq-waste beaker · UV TLC plates + capillaries + chamber (30% EtOAc/hexanes) + small vial · **250-mL Erlenmeyer** · Congo Red paper + stir rod · **125-mL separatory funnel** + ring stand · Erlenmeyer for drying + MgSO_4 · **50-mL beaker + 3 boiling stones** · stemmed funnel + fluted filter paper (gravity) · steam bath + lab jack.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Set up + generate HCl

1. **Weigh a 50-mL RBF** (record). Add **2 mL carvone** w/ a plastic pipette; **weigh again** → exact carvone mass by difference.
2. Add **1 mL H_2O** + a stir bar. Clamp the flask over a stir plate; start stirring.
3. Set up the quench station **before** touching TMSCI: beaker w/ **~75 mL sat. NaHCO_3** + a plastic pipette, and a labeled **aqueous-waste beaker**, both in the hood.
4. Fetch TMSCI: dry **10-mL grad cylinder** inside a **250-mL beaker**, **50-mL beaker inverted on top as a lid** → instructor adds **~6 mL TMSCI** → carry back with **both gloved hands**.

TMSCI is corrosive & fumes. Goggles + gloves. If it touches gloves/clothing, remove them immediately.

5. Add the TMSCI to the stirring carvone/water **through a funnel** (so it all reaches the inside of the flask). If pipetting, hold the pipette **vertical** or it can squirt.
6. **Record the time you finish adding** = t_0 for the reaction.
7. Clean residual TMSCI: pull the sash down; set the funnel on the grad cylinder and rinse both with **sat. NaHCO_3** (it will fizz); park the cylinder at the back of the hood until fizzing stops, then pour the rinse into the aqueous-waste beaker. **Change gloves.**

Part 2 — Monitor by TLC ~20 mL 30% EtOAc/hexanes; one chamber for all plates

8. Spot a **3-lane UV plate**: **S** = carvone standard · **C** = co-spot (S + R in one lane) · **R** = reaction aliquot. *Use a clean capillary for R.*
9. Aliquot = **1 drop** of reaction mixture into a small vial + **2–4 drops eluent**; spot the diluted aliquot.
10. Run the **first plate immediately**; record elapsed time. Repeat TLC **every 15 min** until the carvone spot is gone from R (**15–60 min**). Record the completion time. Used capillaries → glass waste.

Part 3 — Work-up & isolation *this is where product gets lost — follow exactly*

11. Add **5 mL H_2O + 10 mL diethyl ether** to the reaction flask.
12. Transfer everything to a **clamped 250-mL Erlenmeyer**. Pipette in **~5 mL sat. NaHCO_3** ; stir gently. Still fizzing? add more NaHCO_3 in **5-mL portions until fizzing stops**.
13. Confirm neutral w/ **Congo Red paper**: dip a stir rod in the mixture, touch a drop to the strip. **Blue = still strongly acidic** → add more NaHCO_3 and re-test.
14. Transfer to the **125-mL separatory funnel**; let layers separate; **drain off the aqueous (bottom) layer**. *Your product is in the upper ether layer — keep it.*
15. Wash the organic layer with **10 mL aq. NaHCO_3** → let separate → drain aqueous. Then wash with **10 mL brine** (sat. NaCl) → drain.
16. Pour the organic layer into an Erlenmeyer; add a **small scoop of MgSO_4** and swirl. It's dry when some MgSO_4 stays **suspended ("snow-globe")**; if everything clumps at the bottom, add a little more.
17. **Weigh a 50-mL beaker containing 3 boiling stones** — record this mass.
18. **Gravity-filter** the dried ether solution into that weighed beaker (removes MgSO_4). Evaporate the ether on a **steam bath**. *Put the steam bath on a lowered lab jack so condensate drains to the sink.*

Experiment Notes (in lab)

measurement	value
Mass empty 50-mL RBF	g
Mass RBF + carvone	g
Mass carvone	g
Mass beaker + boiling stones	g
Mass beaker + product	g
Mass product	g
% yield	

TMSCI added at (t_0): _____

TLC monitoring

time	R _f carvone (S)	R _f product (R)	carvone gone?
0 min			
15 min			
30 min			
45+ min			

Rxn complete at: _____

Obs: adding TMSCI (heat/fizz): _____

Congo Red result (blue → neutral?): _____

Layers (which drained) / drying: _____

Product appearance · IR peaks: _____

19. When all ether is gone, **cool the beaker, then reweigh** → product mass by difference → % **yield**.

Part 4 — Characterization

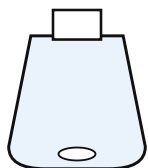
20. IR of **both** your product and carvone. Also finish Act 7: IR, mp & mass of the recrystallized triphenylmethanol.

✍️ WRITE

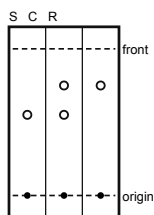
limiting = carvone (mass ÷ 150.22) → theoretical product; % yield = actual ÷ theoretical × 100.

Markovnikov: H⁺ adds to give the **more stable 3° carbocation**; only the electron-rich isolated alkene reacts.

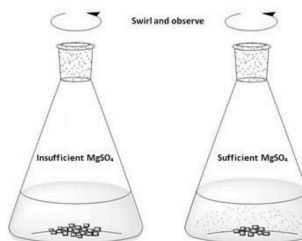
Setup & technique — draw these



50-mL RBF, stirring



TLC monitor (S · C · R)



ion is dry (i.e. has no water remaining), not all of the dry w/ MgSO₄ → “snow-globe”

CONTEXT (don't copy) ▶

Co-spot (C) — S + R in one lane; two separated dots proves the product spot is genuinely new.

Why brine? pulls dissolved water out of the ether layer so MgSO₄ has less to do.

Weighed beaker + stones — you evaporate *into* it, so the final mass minus this = your product.

Clean-up & Conclusions

- Acidic rinses quenched w/ NaHCO₃ → **aqueous waste**; ether/organic filtrates → **organic waste**; used capillaries → glass waste; MgSO₄ + filter paper → solid waste; fresh gloves after TMSCl; bench task.
- Conclusions** (~4 sentences [after lab](#)): TLC evidence carvone was consumed; which alkene reacted & why (Markovnikov, 3° cation, isolated vs conjugated); your % yield + where product was lost; what IR shows vs carvone.